Introduction

The dataset used for this project was the Global Terrorism Database (GTD) containing recorded terrorist attacks from 1970–2020. This dataset was selected because it includes detailed information about incidents such as location, attack type, target type, casualties, and responsible groups. During the project, the focus shifted from predicting the exact date of future attacks to analyzing historical terrorism patterns and identifying the factors most associated with casualties and attack trends. This approach was more realistic and statistically meaningful given the nature of the data.

Objectives

The primary goal of this project was to perform a statistical analysis of global terrorism patterns using regression techniques and exploratory data analysis. The project aimed to determine which variables were most strongly related to the number of fatalities and injuries, while also identifying the regions and countries with the highest concentration of attacks. Another objective was to compare attack methods, target types, and changes in terrorism trends over time. In addition, predictive models were developed to estimate casualties based on characteristics such as attack type, location, and weapon type.

Methodology

First, the dataset was cleaned and prepared in RStudio. Missing values were handled, unnecessary variables were removed, and relevant predictors were selected. Exploratory data analysis was then conducted using summary statistics and visualizations to better understand terrorism trends over time and across regions.

After data preparation, regression models were developed to examine relationships between predictors and casualties. Variables such as attack type, region, target type, and weapon type were tested. Model performance was evaluated using standard statistical measures, and results were interpreted to determine which factors had the greatest impact.

The final report was completed using R Markdown, producing an HTML document. Git and GitHub were also used for version control and project submission.

Expected Outcomes

The completed project was expected to produce an R Markdown report containing the full data analysis process, including data cleaning, preprocessing, visualizations, regression model results, and statistical conclusions. The final outcome was to generate meaningful insights about terrorism trends and the variables most associated with casualties. Success for this project meant demonstrating the ability to apply statistical modeling techniques to a large real-world dataset while clearly communicating the findings through a professional report.

Challenges and Limitations

One challenge was the missing data for the year **1993**, which is a known gap in the GTD dataset. This limited continuity in long-term trend analysis. Another challenge was the large size of the dataset, which required additional processing time and efficient coding practices in R.

Additionally, predicting future terrorist attacks with precision is inherently difficult because terrorism depends on political, social, and unpredictable human factors that historical data alone cannot fully capture. Therefore, the project focused more on explaining patterns than forecasting exact future events.

Conclusion

This project successfully analyzed historical terrorism data to uncover meaningful global patterns and relationships. Rather than predicting the exact date of attacks, the final analysis provided insight into where attacks were most common, what attack methods caused the most harm, and which variables best explained casualties. Overall, the project demonstrated how statistical methods can be used to better understand complex real-world events.

References

Deliverable 1: Project Proposal

Final Project Guidelines